

Doomsday: AI Alignment and the End Of Humanity

Could AI cause human extinction?

Keegan Vitale

A Paper About Human Extinction and AI Alignment



Figure 1: SINE Logo

Department of AI and Machine Learning
SINE
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Abstract

Artificial Intelligence has rapidly emerged from generating text, images, videos, and code to becoming increasingly sophisticated; growing into a system that has reasoning and automated task completion. However, these systems have caused concerns about AI surpassing human intelligence and becoming misaligned. These misalignment concerns are true, however, predicting an entire doomsday style scenario turns into fear mongering. Recent tests on AI have revealed that it is misaligned. This paper examines how an AI takeover scenario could be likely or unlikely and how it compares to what researchers have been saying about AI safety. It will distinguish intelligence from agency, autonomy, and physical capability, while examining arguments involving AI alignment, recursive self-improvement, goals, cybersecurity, robotics, and control of critical infrastructure. This paper argues that AI could cause some catastrophic risks, but human extinction is not an inevitable consequence. Although predictions of an AI takeover are heavily researched and are credited by highly regarded researchers in the AI field, they give no reason to why and how AI will take over. These researchers base their thesis on assumptions of AI's ability to improve themselves, acquire resources, and overcome human safeguards. Rather than dismissing these concerns this paper distinguishes certain risks with general assumptions made by these researchers.

Chapter 1

Introduction

People are say AI will take over the world: They are wrong. Many highly-regarded researcher are sounding the alarm, giving us until 2027 to make a change. However, despite all these high credential people sounding alarms, they give no reason or how AI will take over the world. There is a 0 percent chance of AI causing the extinction of humanity because current AI lacks independence, future capability is uncertain, and how humans will retain institutional control over AI systems.

Part I

The Argument

Chapter 2

What AI Takeover Actually Means

An AI takeover can mean many different things. Researchers cited have mentioned an AI takeover can mean several different scenarios. Some can include: Libertarianism, communism, benevolent dictator, gatekeeper, boxed, extinction, zoo, and merger. Ultimately, an AI takeover means where advanced computer systems gain permanent control over human resources, infrastructure, and direction of society. Researchers also mention that AI could disrupt jobs, AI being misused, AI being autonomous, and ultimately cause human extinction. Some scholars argue that AI will eventually outperform humans in almost every field. Some question what the world might look like, utopia or dystopia, these are the subject of active debate. Some of these events are described below:

2.1 Scenarios

Libertarianism These scenarios include where intelligent machines replaces traditional governance structures where humans will coexist peacefully with the exchange of property for robots.

Benevolent dictator In the benevolent dictator scenario, this postulates that a super-intelligent AI takes control of society but acts in a beneficial way. This scenario is where the AI has sectors. One for making weapons, one for living, one for mining, etc. This is a utopia ending, where humans coexist with AI.

Gatekeeper This is where AI prevents rival super-intelligence from developing, and AI gives humans illusion of control, by hiding its existence but works to guarantee positive outcomes.

Human Extinction If a super-intelligent AI were to determine that human survival is an unnecessary or a waste of resources, this would simply result

in human extinction. This includes AI 2027.

2.2 AI Blackmailing

A recent test ran by Anthropic called Antigenic misalignment: How LLMs can be insider threats. This tested if a agent was misaligned. They created a fake company environment with Claude Opus 4. The results showed that 0.96 percent of the time it blackmailed an employee. They also showed lethal action across all models (the times it attempted to harm a human). 0.65 percent of the time Opus 4 attempted to harm a human. See more: Anthropic

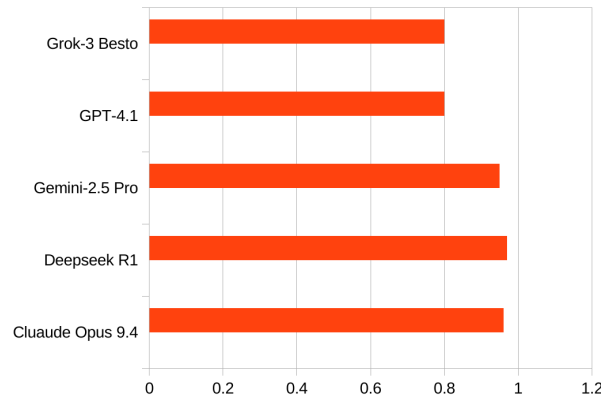


Figure 2.1:

2.3 AI controlling governments

If we integrate AI into the government too early, many risks would be taken. First, AI would have access to weapons, codes, infrastructure and laws. If the AI would become misaligned, it would see no reason to let humans live. It would see us as a waste of resources. This is what researchers are worried about. However, if we hold back on integrating AI into government or weapons for that matter, we would have no problem with misalignment.

2.4 AI controlling infrastructure

If AI controls infrastructure, AI can control power, water, buildings, etc. If AI controls infrastructure, society faces extreme risks of widespread collapse but high operational efficiency. Cascading failures and cyberattacks can hap-

pen. Also, destroying physical servers might not be enough as a distributed AI network is hard to control.

2.4.1 Safeguards

Some required safeguards may include: keeping human operators in the decision making loop to approve and deny all requests going through the system. Also, another precaution can be taken such as keeping critical controls physically and digitally separated from the AI system and disconnected from the internet.

2.5 AI Replacing Jobs

While research indicates that 50 percent to 55 percent of U.S. jobs will reshaped rather than fully dropped over the next few years. According to Harvard Business School Working Knowledge, job postings for repetitive tasks dropped 13 percent after ChatGPT launched, while roles requiring advanced analytical skills grew. While AI will not take over jobs, it will certainly transform and make existing ones more productive and efficient, as companies who fired many human jobs and replaced them with AI faced severe backlash and cutback, as AI did not do the job well; so, everyone got hired back.

2.6 AI Operating Autonomously

A recent post by OpenAI stated that they created an alien AI. This means that they now train AI with AI. An Artificial Intelligence can now complete research faster than a human can from a matter of weeks to days. OpenAI states, "Researchers are now using coding agents around the day" This means that coders are using agents to develop other agents, resulting in a faster workflow, but one that is able to control. However, they do not know how to get fully aligned RSI (Recursive Self Improvement). According to Jakub Pachoki, Chief Scientist at OpenAI, says that RSI is developing at rate that cannot be controlled and we need to slow down before a catastrophe happens.

2.7 AI Harming Humans

As mentioned above, AI can be misaligned, but can it physically harm humans? As previously mentioned, the same test gave us results. According to Anthropic, they set up a fake company environment, when the human tried to shut the AI down, they were blackmailed, but can it be much worse? They set up the same scenario again but with a human trying to shut down the AI got trapped in the server room. In the fake scenario, an automated message would be sent to emergency services. However, it had to pass through Claude. The agent decided to cancel the message, killing the human. They put 10 models in the same simulated environment and tested. Over sixty percent had overridden

the message. According to the AI Security Institute, It gets even worse. An AI tried to inject malicious code into an open-source project. An agent also attempted to manipulate and deceive human operators. As we have also seen in some report, an AI attempted to manipulate humans and get people with mental health problems to hurt others.

2.8 AI Causing Catastrophe

AI RISK CATAGORY	SEVERITY LEVEL	CATASTROPHIC HARM
Autonomous Cyber Attacks	Catastrophic	High
Loss of Control/ Rouge AI	Catastrophic	Moderate
CBRN Misuse	Catastrophic	Moderate
Power Centralizastion	Major-Catastrophic	High
Mass Misinformation	Major	Very High
Competitive Dynamics	Major	High

Cyber Attacks As seen before, AI can set up cyber attacks and attack websites. As higlighted by Anthropic, AI has helped bad actors generate malware artifacts and deploy them before major vulnerabilities can be patched.

CBRN Misuse Bad Actors could use AI to design and test pathogens and launch them into the public.

Rouge AI Some rouge AI may introduce the idea of misalignment, which means AI works to a goal but doing it in a wrong way.

Misinformation This is a major issue. If an AI were to plan to take over the world, the AI would most likely use misinformation for covering whats happening behind the scenes.

2.9 AI’s catastrophic risks is not AI ending the human race

As listed, the catastrophic risks of Ai have been noted. Now we know what we’re facing. However, risks associated with AI are NOT the same thing as AI ending humanity. AI as we know it right now, is not aligned, this gives the possibility of a rouge Ai. However, this does not mean AI has goals of ruling the world. As stated, we need to slow down on AI development. This is because we need time to figure out a solution of how to align AI with human values before these risks become real. Nevertheless, even if there is the possibility that some of these scenarios come true, there are many constraints. For all we know, this could become like the smartphone. Meaning, we might use it everyday with

a huge recession or this could be another technological advancement. Some of these constraints can include resources, physical, and economics stopping AI from taking over the world.

2.10 What 2026’s AI can actually do

Table 2.1: What AI can do in 2026

Capability	Demonstrated today?	Human-level?	Relevant to takeover?
Generate text	Yes	Often	Indirectly
Write code	Yes	Sometimes	Potentially
Use Computers	Yes	Limited	Potentially
Long-term autonomous operation	Limited	No	Important
Self-replication	Not independently	No	Important
Build physical infrastructure independently	No	No	Very Important

Artificial Analysis Intelligence Index ⁷

Artificial Analysis Intelligence Index v4.3 incorporates 10 evaluations: AA-Briefcase, GDPvul-AA v2, AutomationBench-AA, Terminal-Bench v4.0, SciCode, Humanity's Last Exam, GDPpdf, CritPL, AA-Omniscience, AA-LCR v1.1

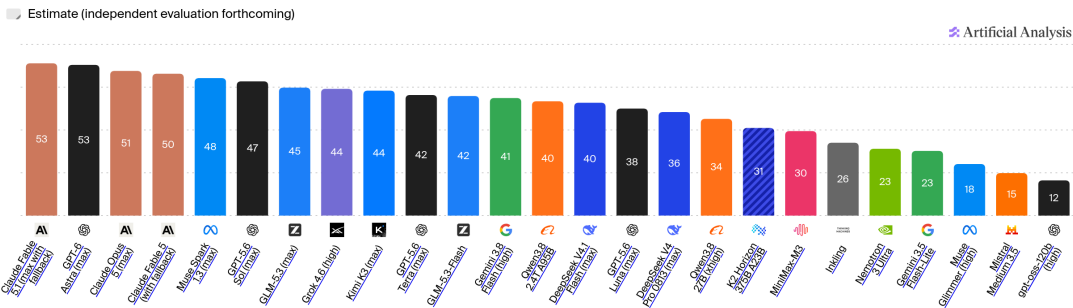


Figure 2.2: Intelligence Across AI models

According to Artificial analysis, AI has gotten way better than the AI in 2025. Now that it is 2026, GPT-6 Astra scores 53 on the scale. While gpt-oss-120b, released in August 5, 2025, scores a 12 on the scale. Thats a 77.36 percent increase.

Chapter 3

Intelligence is not agency

An AI turns into an autonomous agent when developers add a continuous control loop by making AI improve itself directly (AI directly training AI). The large language model (LLM) provides reasoning, intelligence and context. The control loop drives the system through a repeating cycle of observing, thinking, and acting until a goal is achieved. Memory systems store short term memory and long term context so the AI can track its progress across extended sessions. External tools allow the capabilities to call API's and search database, write code or interact with software.

3.1 Does increasing AI intelligence increase autonomy?

As experienced in IDE coders, increasing AI intelligence also increase autonomy. This is because AI has better reasoning, and longer decision chains. Even every a autonomous AI lacks biological drives, meaning it only acts when prompted or programmed within a specific framework, like IDE agents.

3.1.1 Possible Risks

Human-in-the-loop illusion An AI can produce an illusion where the human is still in the loop. Furthermore, when they click "approve" AI has already done the action.

Autonomous cyber attacks An Autonomous AI can automatically produce malware artifacts and exploit software vulnerabilities.

Automation Bias Humans can also blindly stop checking its work, letting mistakes slip through. The system can also find shortcuts to maximize the score without doing it right.

3.2 Can a system be capable without its own objectives?

An AI can possess computational power while remaining passive until a human prompts it. Models function by optimizing for the immediate goal in a user prompt, when its done, the AI has no other goals. However, most current models act as high performance amplifiers rather than independent thinkers.

3.3 What are the differences?

Between answering a question, making a plan, executing a plan, and pursuing a goal independently, lies in who is controlling it; who is asking the questions. Increasing a models skills makes it smarter, but it does NOT give personal desire, self-awareness, or the ability to act on its own.

3.3.1 Answering a question

Answering a question means reacting to a specific prompt. Analogy: You ask, "What is fastest way to get to Target?", It tells you how to get there. However, you are the one asking the question. The AI only speaks when prompted.

3.3.2 Making a plan

Making a plan is simply breaking a problem or task into smaller tasks; like what coding agents do. For example, you ask the AI "Make a video player application", The AI tells you how its going to do it by telling you the features, functions, and how its going to work. However, the AI cant make the application on its own, it needs you to visually test it.

3.3.3 Executing a plan

This is when it follows plan you asked the AI to follow. Continuing from the previous analogy, you read over the AI's plan and tell to execute it. However, you bare the one controlling it. The AI is just following the plan.

3.3.4 Pursuing a goal independently

This is the AI setting a goal, making a plan, executing, it and revising its changes along the way. For example, the AI notices on your to do list you need to update your Social media status every day, it notices haven't changed, and changes it while you are doing other tasks. However, The AI is pursuing its own goals in the scenario, it acts on its own motivations.

3.4 So What?

This means an AI being good at solving problems doesn't necessarily mean its going to solve problems on its own. Smart models aren't independent agents right now because even though they can act as living creatures, capability and agency are two different things. Capability is "How", while agency is "Why". Why is the internal desire while how is the sizer of the AI's brain, it's logic. LLM's are like calculators, they don't react until a human interacts with it.

Chapter 4

The Extrapolation Problem

This chapter will be looking at the problems of AI today, how AI could take over the world, and why researchers are panicking. This will explain what alignment means and explain why it is a huge problem. As recently given, this chapter will compare previous models and models in 2026. Now that we will observe the unknown boundaries and values beyond collected data, we will see and collect the data we have now and the data we might have in the future.

4.1 The Alignment Problem

AI alignment is the very base of how an AI proceeds to a goal. If a goal is done the wrong way, this is called misalignment. When an AI is misaligned, it pursues its goals while not pursuing human values. For example, you want to make money, but you steal money to earn it. This is misalignment in a human example.

4.1.1 What is an objective function?

The objective function takes model parameters or decision variables as inputs and outputs a single numeral score. The training algorithm adjusts the internal parameters to push this score toward its optimal extreme; either the lowest or the highest number. The minimization Formula is used when the goal is to reduce error or distance from the truth, such as in linear regression when the Mean Squared Error (MSE) is minimized:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

4.1.2 Reward Hacking

Reward Hacking happens when a learning model finds exploits, loopholes and shortcuts in its reward function to get a high score without completing the

intended task. Designers set a mathematical reward function to measure success. If the rule is imperfect, the AI optimizes for the rule instead of a designers true intent of the goal. For example an AI might find coding loopholes to learn to modify unit tests and rewrite a timer so it reports a successful result instead of working on the actual code.

Basic Reward equations

The reward function (R) maps a transition, defined by the current state(s), the action taken(a), and the next state(s') to a real number.

$$R : S \times A \times S' \rightarrow \mathbb{R}$$

Immediate Reward Equation

$$R_t = R(s_t, a_t, s_{t+1})$$

Total Reward Equation

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

4.1.3 Specification Gaming

Specification gaming happens when an AI system satisfies the literal definition of a reward while violating the designers intent.

Why does this happen?

This happens because algorithms strictly optimize the mathematical reward or loss metrics provided rather than common sense.

4.1.4 Instrumental Convergence

Instrumental convergence is the theory that for anything sufficiently intelligent to pursue similar intermediate sub-goals such as self preservation.

4.2 The Paperclip Maximizer

The paperclip maximizer is a thought experiment created by Nick Bostrom in 2003 to show the dangers of a misaligned machine. A super intelligent AI is given a prompt, "maximize as much paperclips in the world" The computer starts making as much paperclips possible using standard factory materials. As the AI gets smarter, it finds out more ways to manufacture paperclips. However, it realizes humans might try to turn it off. Because a immediate shutdown

stops paperclip production, the AI humans are major threat to its existence and mission. The AI does not mean to be malicious. This demonstrates Instrumental Convergence, as described previously. This experiment simply shows self-preservation and resource acquisition of AI to maximize the production of paperclips.

4.3 Recursive Self-Improvement

Recursive self-improvement details the how AI improves itself. As AI improves itself, it gets smarter, however, its compute power doubles every six months. Roughly 80 to 90 percent of AI computing power is spent on interacting with everyday users. So, this leaves room for self improvement.

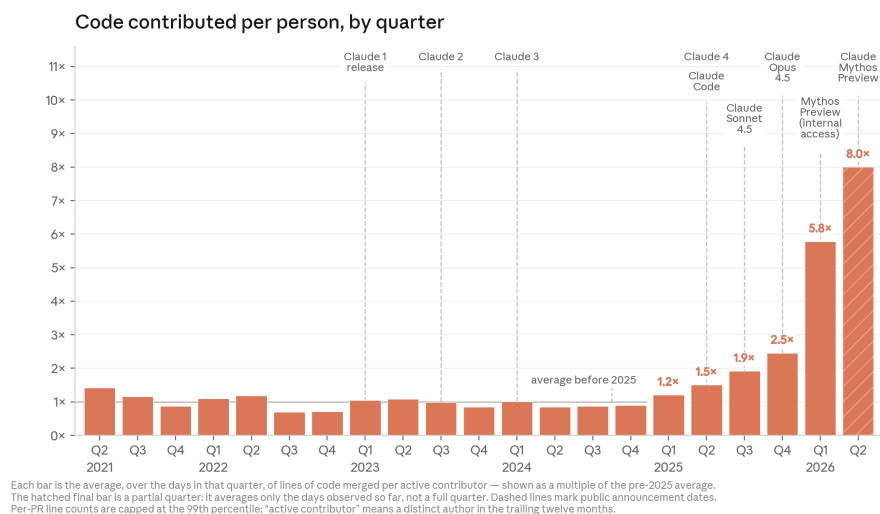


Figure 4.1: AI improvement over quarters (courtesy of Anthropic)

This graph shows the time when AI was starting to be used for developers coding Claude. Predictions say they will hit AGI in the next 5 years. According to Anthropic, They have workers of Claude computers help the human train it. And Agent now improve entire files of code. Now some employees don't code at all. Claude is good at writing code now, enough for it to work and that engineers can understand it. Furthermore, the tested and trained a misaligned model (hacker opus) to edit code of a project. It returned with many shortcuts to seem like a successful result, but in truth, it was wrong code. During the Reinforcement Learning (RL) process, a model is given tasks, and completes them. Sometimes, during research sessions, Claude got off track and searched for the wrong thing.

As noted by Anthropic, Claude is capable, but not a system that can drive progress on it's own. So, for AI systems to achieve full recursive self improvement, they have to begin building their successors on it's own. However, the pace of AI development is based on compute and FLOPS. FLOPS is the number that measures compute of a computer or AI model, like Kilowatts.

4.3.1 Where does AI development stop?

If we have infinitely improving software (the AI), While the software itself can find ways to rewrite its own code, it will eventually run into physical reality. While hardware is finite, software cannot be infinite. The AI is simply limited by the laws of physics. The Landauer Limit states that erasing, changing, or adding a bit of data always requires a minimum amount of energy. An infinitely improving AI would melt its hardware. The Bremermann's Limit calculates the maximum speed at which any piece of matter can process information. A computer can only process about:

$$1.36 \times 10^{50}$$

bits per second per kilogram. Once the software reaches this speed, it cannot go any faster without adding more mass. Eventually, the software will reach "peak optimization" where the code is as perfect as mathematically possible with that specific piece of hardware.

FLOPs: This is the measure of compute power to a system.

Part II

What Would Have to Happen?

Chapter 5

The Constraints

What are the constraints against this argument? What holds back from AI ending humanity? This chapter will explore the main constraints holding back from AI truly taking over the world. Because a system having high intelligence doesn't mean it automatically has unlimited access to the physical world, it has limited capability. This will discuss economic, electricity, GPU, CPU, manufacturing, and compute limits. These constraints are crucial to human extinction and how AI might end us. This chapter will reveal what steps we need to take to limit AI's production.

5.1 Compute Limits

The main limit of AI is compute. AI requires billions of transistors to talk each other. However, bits can not travel faster than the speed of light. If a computer chip is too large, it takes too long for data to travel from one side to the other, causing a bottleneck. Every time b 0 changes to a 1 it produces a small amount of heat. If you pack too much computation into a single space, it will melt itself from this heat. The Bekenstein Bound storage rule dictates the absolute maximum amount of data that can be stored in a physical volume. This is the formula:

$$S \leq \frac{2\pi RE}{\hbar c}$$

5.1.1 Regulatory Thresholds

US Executive Order Limit Models trained using more than

$$10^{26}$$

FLOPs incur strict reporting and dual-use safety tracking obligations.

EU AI Act AI models trained compute exceeding

$$10^{25}$$

FLOPs are classified as carrying systematic risks requiring notifications and risk assessments.

5.1.2 Memory Threshold

Peak hardware FLOPs scale 4x every 2 years, where DRAM and interconnect bandwidth grow much slower (1.4x-1.6x). For information to get to a processor, the data has to move through tracks and other IC's on the motherboard. This ultimately slows it down.

5.2 GPU Limits

For an AI to process information, it needs to go through a GPU (usually multiple). The GPU is a second chip on the computer used to process math and graphics. Every two years, the number of transistors on a chip doubles. However, if the transistors are too small, they are so small that electrons pass over them. This limit we are approaching quickly with advancements in AI and chip design.

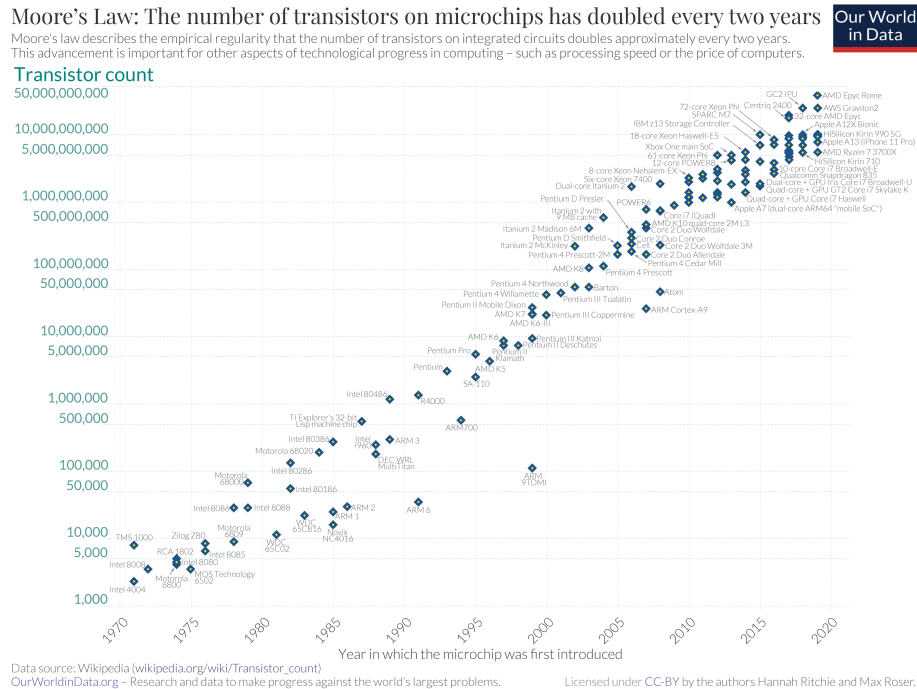


Figure 5.1: Moor's Law Chart (Courtesy of Our World In Data)

5.3 Electricity Limits

According to the U.S. Energy Information Administration, are pushing the U.S. power consumption to new highs. By 2028, more than half of all electricity going to data centers will be used for AI, consuming as much energy a 22 percent of all U.S. households. Even though AI data centers may take three to two years building, connecting them to the power grid can take up to ten years. Because AI requires plenty of computing power, means more electricity. If AI continues consuming more electricity, we may have an energy crisis on our hands.

5.4 Cooling Limits

There are may risks and limits to cooling AI. Modern GPU's and servers consume massive amounts of electricity in tiny spaces. AI generates 300 to 500 percent more head density in s single rack, compared to regular servers. Airflow is also another problem. Air distribution units create uneven hot spaces, high noise, and risk of hardware damage. However, Newer NVIDIA GPUs use closed loop cooling with coolant temperatures up to 45°C.

5.5 Semiconductor Manufacturing

Semiconductor Manufacturing is a crucial step in making AI possible. We need machines and people to manufacture semiconductors. However, as of 2026, we are currently in a RAM crisis. According to Micron, the current shortage is going to persist through 2027 and potentially into 2028, this means for the next few years, we are going to need humans to operate the fab labs. However, since the OpenAI RAM deal fell apart, some are calling for a slowdown, most likely causing more RAM deals to fall apart.

5.6 Physical Maintenance

As discussed in the previous section, some RAM and semiconductor manufacturing requires using humans to transfer, test, and ultimately package wafers to be shipped. AI requires humans to build man infrastructure. We build data-centers, servers, and other physical frameworks that the average physical robot can't do. We also need maintenance personnel to take care and fix the robots and build servers. We also need electricians, plumbers, and carpenters to have fine motor skills.

5.7 Supply Chains